

Laporan Tabel Analitik Sepak Bola Internasional (1872-2026)

Dokumen ini memuat seluruh data dari laporan analisis sejarah sepak bola internasional yang telah dikonversi menjadi format tabel profesional untuk meningkatkan keterbacaan dan struktur laporan.

2. Data Understanding

Gambaran Umum Dataset

Dataset	Rows	Columns	Duplicate Rows	Missing Values	Memory MB
results	49,502	9	0	14	17.64
shootouts	681	5	0	422	0.18
goalscorers	47,867	8	79	298	14.08
former_names	36	4	0	0	0.01

Nilai Hilang yang Penting (Important Missing Values)

Dataset	Column	Missing	Missing %
results	home_score	7	0.01%
results	away_score	7	0.01%
shootouts	first_shooter	422	62.0%
goalscorers	scorer	44	0.09%
goalscorers	minute	254	0.53%

3. Data Cleaning

Aksi Pembersihan Data (Cleaning Actions)

Task	Result
Convert date fields	Dates converted in all relevant files
Remove duplicates	79 duplicate goalscorer rows removed; no duplicate result rows
Check impossible scores	No impossible score rows found after excluding missing scores
Handle missing score rows	7 result rows with missing scores removed from match analysis
Standardize text	Team, tournament, city, country, scorer, winner fields stripped and normalized
Convert TRUE/FALSE	neutral, own_goal, and penalty converted to boolean fields
Check missing shootout winner	No missing shootout winners
Check missing first shooter	422 missing first-shooter records retained as unknown

4. Feature Engineering

Fitur yang Dibuat (Created Features)

Feature	Description
match_result	Home Win, Draw, or Away Win
total_goals	home_score + away_score
goal_difference	home_score - away_score
year, month, decade	Time-based features
scoreline	Combined score, such as 1-0 or 2-1
tournament_category	Friendly, World Cup, Continental Cup, Qualifier, Other
home_advantage_flag	True when match is not neutral
previous win-rate features	Historical home/away win rates before each match
previous goal-difference features	Historical average goal difference before each

Feature	Description
	match
recent form	Last 5 and last 10 match form
ELO rating	Rolling pre-match ELO-style team strength

5. Dataset Overview Findings

Metrik Utama

Metric	Value
Cleaned matches	49,495
National teams	336
Tournaments	200
Host countries	269
Host cities	2,089
Total goals	145,471
Average goals per match	2.94
Date range	1872-11-30 to 2026-07-04

Distribusi Hasil Pertandingan (Match Outcome Distribution)

Outcome	Matches	Share
Home Win	24,256	49.0%
Away Win	13,983	28.3%
Draw	11,256	22.7%

6. Home Advantage Analysis

Keunggulan Tuan Rumah Berdasarkan Kategori Turnamen

Tournament Category	Matches	Home Win %	Draw %
World Cup	132	62.1%	17.4%
Continental Cup	1,988	52.8%	21.3%
Other	4,250	52.6%	21.2%
Qualifier	14,264	52.3%	21.3%
Friendly	15,724	48.5%	25.0%

7. Goal Analysis

Skor Paling Umum (Most Common Scorelines)

Scoreline	Matches
1-0	5,104
1-1	4,920
0-0	3,972
2-0	3,841
2-1	3,780
0-1	3,462
1-2	2,552
3-0	2,370
0-2	2,229
2-2	1,945

Rata-rata Gol Berdasarkan Kategori Turnamen

Category	Matches	Avg Goals	Draw %
Other	9,633	3.14	21.0%
Continental Cup	4,492	3.05	22.9%

Category	Matches	Avg Goals	Draw %
Friendly	18,388	2.87	25.1%
Qualifier	15,928	2.87	21.0%
World Cup	1,054	2.83	22.5%

8. Team Performance

Tim dengan Persentase Kemenangan Tertinggi (Minimal 100 Pertandingan)

Team	Matches	Win %	Goals For	Goals Against	Goal Difference
Jersey	235	65.1%	645	285	360
Brazil	1,063	63.5%	2,314	959	1,355
Guernsey	240	60.4%	616	313	303
Spain	787	59.0%	1,610	703	907
Germany	1,035	58.1%	2,331	1,203	1,128
England	1,094	57.4%	2,389	1,044	1,345
Iran	615	56.7%	1,158	486	672
Argentina	1,073	55.5%	2,039	1,078	961

Tim dengan Jumlah Pertandingan Terbanyak

Team	Matches
Sweden	1,105
England	1,094
Argentina	1,073
Brazil	1,063
Germany	1,035
South Korea	1,010

Team	Matches
Mexico	1,007
Hungary	1,006

Selisih Gol Terbaik (Minimal 100 Pertandingan)

Team	Goal Difference
Brazil	1,355
England	1,345
Germany	1,128
Argentina	961
Spain	907
South Korea	877
Netherlands	771
Sweden	747

Rekor Pertahanan Terbaik Berdasarkan Rata-rata Kebobolan

Team	Avg Goals Against
Iran	0.79
Morocco	0.80
Spain	0.89
Brazil	0.90
South Korea	0.91

9. Tournament Analysis

Turnamen dengan Pertandingan Terbanyak

Tournament	Matches	Avg Goals	Draw %
Friendly	18,388	2.87	25.1%
FIFA World Cup qualification	8,771	2.89	21.1%
UEFA Euro qualification	2,824	2.83	19.6%
African Cup of Nations qualification	2,327	2.42	24.5%
FIFA World Cup	1,054	2.83	22.5%
Copa America	869	3.14	21.6%
African Cup of Nations	845	2.38	28.6%
AFC Asian Cup qualification	829	3.29	16.5%

10. Country and City Hosting

Negara Penyelenggara Pertandingan Terbanyak

Country	Matches
United States	1,573
France	932
Malaysia	830
England	773
Qatar	760
Thailand	723
Germany	685
Sweden	654
South Africa	611
Spain	609

Kota Penyelenggara Pertandingan Terbanyak

City	Matches
Kuala Lumpur	745
Bangkok	580
Doha	560
London	453
Budapest	437
Kuwait City	377
Montevideo	375
Vienna	365
Oslo	359
Glasgow	356

11. Time Series and Historical Evolution

Tren Dekade Terpilih (Selected Decade Trends)

Decade	Matches	Avg Goals	Participating Teams
1870s	13	4.54	3
1900s	138	4.16	23
1930s	1,079	4.32	83
1960s	2,971	3.48	161
1970s	4,133	2.97	199
1980s	5,025	2.53	194

12. Head-to-Head Analysis

Rivalitas Paling Sering Terjadi (Most Frequent Rivalries)

Rivalry	Matches	Avg Goals	First Match	Last Match
Argentina vs Uruguay	183	2.74	1902-07-20	2025-03-21
Austria vs Hungary	137	4.02	1902-10-12	2016-06-14
Belgium vs Netherlands	129	3.92	1905-04-30	2022-09-25
Guernsey vs Jersey	119	3.45	1905-04-27	2022-05-21
England vs Scotland	118	3.27	1872-11-30	2023-09-12
Argentina vs Brazil	110	3.00	1914-09-20	2025-03-25

13. Penalty Shootout Analysis

Dataset Adu Penalti

Metric	Value
Shootouts	681
Known first-shooter records	259
First shooter win rate	52.5%

Tim Adu Penalti Terbaik (Minimal 5 Kali Adu Penalti)

Team	Shootouts	Wins	Win %
Padania	6	6	100.0%
Indonesia	11	10	90.9%
Ethiopia	8	7	87.5%
Guinea	10	8	80.0%

Team	Shootouts	Wins	Win %
Guatemala	5	4	80.0%
Oman	5	4	80.0%
Martinique	9	7	77.8%
Saudi Arabia	9	7	77.8%
Iraq	15	11	73.3%
Panama	7	5	71.4%

14. Goalscorer Analysis

Pencetak Gol Terbanyak (Top Goalscorers)

Player	Goals
Cristiano Ronaldo	124
Harry Kane	74
Lionel Messi	70
Robert Lewandowski	68
Romelu Lukaku	66
Edin Dzeko	58
Aleksandar Mitrovic	52
Kylian Mbappe	52
Luis Suarez	51
Erling Haaland	51

Pencetak Gol Penalti Terbanyak (Top Penalty Scorers)

Player	Penalty Goals
Harry Kane	23
Cristiano Ronaldo	22

Player	Penalty Goals
Lionel Messi	14
Robert Lewandowski	14
Hristo Stoichkov	13
Kylian Mbappe	11
Cuauhtemoc Blanco	11
Mile Jedinak	11

Tim yang Paling Banyak Diuntungkan oleh Gol Bunuh Diri

Team	Own Goals Benefited
England	26
Spain	26
France	24
Germany	21
Russia	19
Portugal	19
Italy	19
Belgium	18
Brazil	18
Mexico	17

16. Statistical Analysis

Hasil Pengujian Statistik

Test	Statistic	Finding
Chi-square: venue type vs outcome	251.93	Home/neutral context is associated with outcome distribution

Test	Statistic	Finding
Welch t-test: non-neutral vs neutral goal difference	14.19	Non-neutral matches have a higher average home goal difference
ANOVA: average goals by tournament category	35.65	Average goals vary by tournament category
Pearson correlation: year vs total goals	-0.180	Goal scoring declines over time in linear terms
Spearman correlation: year vs total goals	-0.129	Rank trend also confirms lower scoring in more recent eras

18. Machine Learning

Grup Fitur Utama Model (Main Model Features)

Feature Group	Examples
Recent form	Home form last 5, away form last 5
Historical strength	Previous win rate, previous average goal difference
ELO rating	Home ELO, away ELO, ELO difference
Match context	Neutral venue, tournament, tournament category
Time	Year

Hasil Model (Model Results)

Model	Accuracy	Macro F1
Majority baseline	47.7%	21.5%
ELO + form multinomial logistic	59.2%	43.2%

Kepentingan Fitur (Feature Importance)

Feature	Importance
ELO difference	0.379
Away previous average goal difference	0.134
Home previous average goal difference	0.108
Neutral venue	0.101
Home form last 5	0.047
Away form last 5	0.042
Home previous win rate	0.042
Away previous win rate	0.040

19. Model Interpretation and Error Analysis

Sumber Kesalahan Umum Model (Common Sources of Model Error)

Error Source	Explanation
Draw prediction difficulty	Draws are inherently less separable from narrow home or away wins
Squad and player context missing	Injuries, lineups, managers, rest days, and travel are not included
Tournament incentives	Knockout, group-stage, and friendly incentives differ
Historical imbalance	Early football and modern football have different scoring environments
Team-name/geopolitical shifts	Historical continuity can be complex across former names

21. Deliverables Produced

Hasil yang Diproduksi (Deliverables)

Deliverable	Description
Clean dataset	Cleaned match, shootout, goalscorer, and feature-engineered tables
Feature-engineered dataset	Match result, goals, date, tournament category, form, ELO, and historical features
EDA report	This report plus workbook tables and charts
Statistical findings	Chi-square, t-test, ANOVA, Pearson, Spearman
Dashboard guidance	Power BI page plan and export-ready tables
Prediction model	Baseline and ELO/form model
Model evaluation	Accuracy, macro F1, feature importance, error-analysis notes